

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0374

Clover 345 kV
March 24, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0374. The Interconnection Customer proposes a 295.1 MW natural gas combined cycle facility interconnecting at the Clover 345 kV substation in the Virginia area. The request was received on August 30, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 196 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,962,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CHANCELLOR ROW - BER ISLAND 345 KV L	BIRCHWOOD - LEAVITT 345 KV	108.3	1,105	19.2
DIXLETA - HUGHSON 345 KV	DIXLETA - RUSTIC 345 KV L	107.8	1,350	18.6
CARRABELLE - LOUISA GENERATION FACILITY 345 KV	N ANNA - LOUISA GENERATION FACILITY 345 KV	105.9	1,562	11.5
RUSTIC - DIXLETA 345 KV	HUGHSON - DIXLETA 345 KV	105.4	1,314	16.9
BEAR ISLAND - DUNN SIDING 345 KV Line	LEAVITT - BIRCHWOOD 345 KV	104.4	1,570	19.3
BICHWOOD - LEAVITT SUBSTATION 345 KV	BEAR ISLAND - DUNN SIDING 345 KV Line	101.8	1,748	22.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHANCELLOR ROW - BER ISLAND 345 KV L	Reconductor line	\$12,533,000	9% (\$1,113,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DIXLETA - HUGHSON 345 KV	Reconductor line	\$9,511,000	6% (\$553,000)
LOUISA GENERATION FACILITY - CARRABELLE 345 KV	Reconductor line	\$50,332,000	6% (\$2,892,000)
RUSTIC - DIXLETA 345 KV	Reconductor line	\$32,905,000	6% (\$1,905,000)
BEAR ISLAND SUBSTATION - DUNN SIDING 345 KV	Reconductor line	\$16,493,000	8% (\$1,303,000)
LEAVITT - BIRCHWOOD SUBSTATION 345 KV	Reconductor line	\$38,315,000	8% (\$3,196,000)
	Total allocated to project		\$10,962,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.